

Regression Analysis (II) Solutions to Final Exam.

Dec. 16, 2019

- 1.(190) Assume that $Y \sim \Gamma(1, \mu)$, i.e., the pdf of Y is $f(y, \mu) = \frac{1}{\mu} e^{-y/\mu} I(y > 0)$, $\mu > 0$.
- (1)(30) Show that the distribution of Y belongs to an exponential family with the natural parameter θ , and derive θ . (Sol.) $f(y, \mu) = \exp[y(-\frac{1}{\mu}) - \log \mu]$, so that $\theta = -1/\mu$.
- (2)(30) Derive the natural link function. (Sol.) $b(\theta) = \log \mu = -\log(-\theta)$.
- (3)(30) Show that the variance function is $V(\mu) = \mu^2$. (Sol.) $b'(\theta) = -1/\theta \Rightarrow b''(\theta) = \theta^{-2} \Rightarrow V(\mu) = \theta^{-2} = (-\mu^{-1})^{-2} = \mu^2$.
- (4)(50) Let $Y_1, \dots, Y_n$ be independent and $Y_i \sim \Gamma(1, \mu_i)$, $i = 1, \dots, n$. When we fit to the generalized linear model with canonical link, derive the deviance. (Sol.) $l(\hat{\beta}_{max}; \mathbf{y}) = \sum(-\frac{y_i}{\hat{\mu}_i} - \log y_i)$ and $l(\hat{\beta}; \mathbf{y}) = \sum(-\frac{y_i}{\hat{\mu}_i} - \log \hat{\mu}_i)$, so that $D = 2 \sum[-\log(y_i/\hat{\mu}_i) + (y_i - \hat{\mu}_i)/\hat{\mu}_i]$.
- (5)(50) Under (4), derive the Anscombe residual for y_i . (Sol.) $A(y) = \int V(y)^{-1/3} dy = \int y^{-2/3} dy = 3y^{1/3}$. Now, $V(A(Y)) \simeq A'(\mu)^2 V(\mu) = \mu^{2/3}$, so that the Anscombe residual becomes $3(y_i^{1/3} - \hat{\mu}_i^{1/3})/\hat{\mu}_i^{1/3}$.
- 2.(50) Consider a logistic regression model $\text{logit}(\pi) = \beta_0 + \beta_1 X$, and assume that the response $Y_i \sim B(m_i, \pi_i)$, $i = 1, 2, \dots, n$. Assume that $\hat{\beta}_0 = -1$, $\hat{\beta}_1 = 1$, $\text{Var}[\hat{\beta}_0] = \text{Var}[\hat{\beta}_1] = 1$, $\text{Cov}[\hat{\beta}_0, \hat{\beta}_1] = 1/2$. Find the point estimate of x_m , where $e^{\hat{\beta}_0 + \hat{\beta}_1 x_m} / [1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x_m}] = 0.5$. Further, find $100(1 - \alpha)\%$ approximate C.I. for x_m . (Sol.) $\hat{x}_m = -\hat{\beta}_0/\hat{\beta}_1 = 1$. Now, $\text{Var}(\hat{x}_m) \simeq (1 + 2(1/2) + 1(1))/1 = 3$ (see text p. 337), so that $100(1 - \alpha)\%$ approximate C.I. for x_m is $1 \pm z_{\alpha/2} \sqrt{3}$.
- 3.(80) Consider a oneway ANOVA model with replications s.t. $E(Y_{ij}) = \mu + \alpha_i$, $i = 1, 2, 3$; $j = 1, 2$, and we observed $y_{11} = 9$, $y_{12} = 4$, $y_{21} = 10$, $y_{22} = 5$, $y_{31} = 11$, $y_{32} = 6$.
- (1)(50) When we write the model in matrix form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, compute the rank of $\mathbf{X}^T \mathbf{X}$. Also, compute the fitted value for y_{22} under sum-to-zero constraint. (Sol.) Can show that rank of $\mathbf{X}^T \mathbf{X}$ is 3, and $(\hat{\mu}, \hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3) = (7.5, -1.0, 0.0, 1.0)$. Hence, $\hat{y}_{22} = \hat{\mu} + \hat{\alpha}_2 = 7.5$.

(2)(30) Under corner point restriction, compute the fitted value for y_{22} , and compare with the result in (1). (Sol.) Under corner point restriction $\alpha_1 = 0$, $(\hat{\mu}, \hat{\alpha}_2, \hat{\alpha}_3) = (6.5, 1.0, 2.0)$. Hence, $\hat{y}_{22} = \hat{\mu} + \hat{\alpha}_2 = 7.5$. Both estimates under sum-to-zero constraint and corner point restriction should be the same.

4.(80) Let Y and Z be two r.v's.

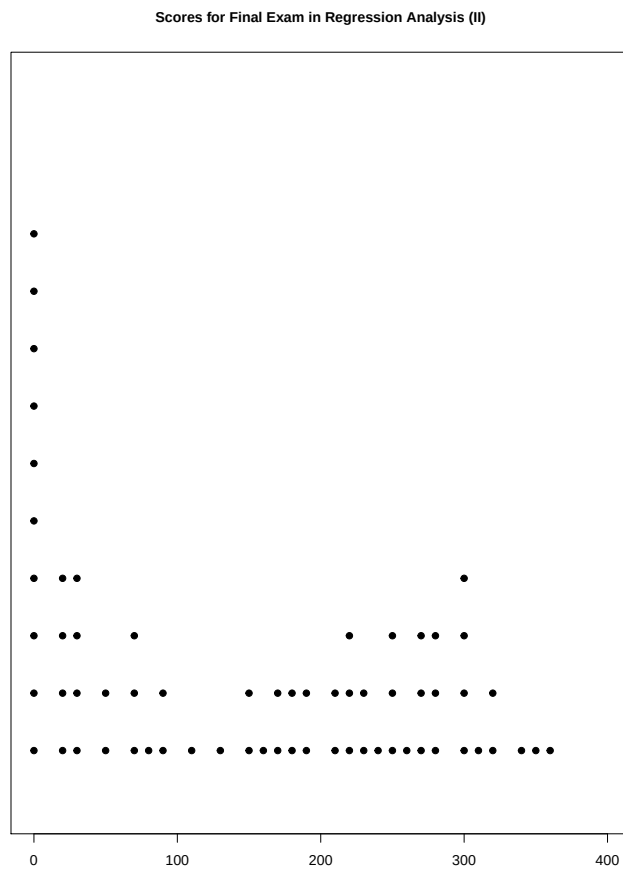
(1)(30) Show that $Var(Y) = E[Var(Y|Z)] + Var[E(Y|Z)]$. (Sol.) Done in class.

(2)(50) Assume that the conditional distribution of Y given Z is Poisson with mean Z , and the distribution of Z is $\Gamma(\theta\mu, \theta^{-1})$, i.e., $f(z) = \theta^\theta \mu^\theta z^{\theta\mu-1} e^{-\theta z} / \Gamma(\theta\mu)$. Using the result of (1), compute $E(Y)$ and $Var(Y)$. Further, interpret overdispersion using this result. (Sol.) First, by the double expectation theorem, $E(Y) = E[E(Y|Z)] = E(Z) = \theta\mu\theta^{-1} = \mu$. Second, by result of (1), $Var(Y) = E[Var(Y|Z)] + Var[E(Y|Z)] = E(Z) + Var(Z) = \mu(1 + \theta^{-1})$. Therefore, if θ is small, $Var(Y)$ will be much larger than $E(Y)$, so that overdispersion occurs.

Summary Statistics for Final Exam. Scores

$n = 65$, $Ave. = 164$

Top Seat : 박혜림



Summary Statistics for Total Scores

$n = 64$, $Ave. = 455$

Top Seat : 박혜림

